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Gauge Fiber Bundle Geometry of Transformers

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Abstract

Building on a companion NeurReps paper that *completely characterizes* the head-wise gauge symmetries of multi-head attention [21], we take the maximal head-wise symmetry group as given and study the induced geometry on the quotient of functionally distinct models. On a generic regular set of parameters, this symmetry group acts freely and properly, so the parameter space fibers over a quotient manifold with gauge orbits as fibers. We define a canonical Euclidean Ehresmann connection, which resolves the degeneracy of the Fisher–Rao (FR) metric along gauge directions; the natural gradient is then the *horizontal Riesz representative* of the Euclidean gradient with respect to the FR geometry on the quotient. The connection has generically nonzero curvature, implying path-dependent holonomy in parameter updates. MHA and FFN blocks separate geometrically: MHA parameters carry gauge symmetry while FFN gradients are strictly horizontal, as the FFN parameters are invariant under the MHA gauge group. A gauge-aware gradient split and a small-loop holonomy estimator give computable diagnostics; Euclidean-proxy consistency checks are consistent with the theory. RoPE restricts the Q/K gauge group to a commutant subgroup, reducing the per-head gauge dimension from d_k^2 to d_k.

1. Introduction

Standard multi–head self–attention layers admit systematic head–wise changes of basis on the key/query and value channels, together with permutations of heads, that leave the realized function unchanged. In this paper we analyze the resulting gauge group, the induced quotient manifold of functionally distinct parameters, and learning dynamics on that quotient.

Preliminaries. Let $\pi : \Theta_0 \to \mathcal{Q}$ denote the parameter–function map on a regular (Zariski–open) subset Θ_0 of parameters. The fibers of π are gauge orbits of a Lie group G. The tangent space $T_\theta\Theta$ admits a decomposition $T_\theta\Theta = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$ into vertical directions $\mathcal{V}_\theta = \ker d\pi_\theta$ and horizontal directions $\mathcal{H}_\theta$. An Ehresmann connection specifies $\mathcal{H}_\theta$ smoothly. Because the empirical Fisher (Gauss–Newton) metric g_θ is degenerate along $\mathcal{V}_\theta$, we use the ambient Euclidean inner product to define the canonical Euclidean connection via $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\mathrm{Euc}}}$. Under a mild Fisher-regularity assumption on the data (Assumption 1), the only degeneracies of g_θ come from gauge directions, so its restriction to $\mathcal{H}_\theta$ is positive definite and induces a Riemannian geometry on the quotient $\mathcal{Q}$ by identifying horizontal vectors along gauge orbits.

Companion paper and scope. A separate companion paper [21] provides a complete classification of gauge symmetries in Transformer architectures. It proves that, on a generic stratum satisfying mild rank and distinctness conditions, the maximal head-wise gauge group for a standard multi-head attention (MHA) block is

$$G_{\max} = ((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h) \rtimes S_h,$$

and develops variants for RoPE, grouped/multi–query attention, LayerNorm placement, and layer-wise factorization. In this *geometry-focused* paper we take those symmetry results as given, recall only the minimal statements we need (Section 2), and concentrate on the induced principal-bundle structure, canonical connection, curvature, diagnostics, and optimization implications.

This paper makes the following contributions:

- **Principal-bundle geometry.** On the generic stratum Θ_0 (Definition 1), $G_{\max}$ acts freely and properly, so $\pi : \Theta_0 \to \mathcal{Q} := \Theta_0/G_{\max}$ is a principal $G_{\max}$–bundle (Theorem 3).
- **Canonical connection and quotient Fisher geometry.** The canonical Euclidean connection $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\mathrm{Euc}}}$ (Section 3) makes the empirical Fisher metric nondegenerate on the quotient, and the natural gradient is the horizontal Riesz representative of the Euclidean gradient (Theorem 7).
- **Curvature, holonomy, and path dependence.** The canonical connection has generically nonzero curvature (Theorem 8), leading to measurable small-loop holonomy and path-dependent training dynamics (Theorem 10).
- **MHA/FFN separation.** Gauge symmetry is localized to the MHA block; FFN parameters lie entirely in the horizontal, function-changing subspace (Proposition 9).
- **Diagnostics and empirical checks.** Gauge-aware gradient splitting and a small-loop holonomy estimator (Section 5) turn these constructions into computable procedures; Section 7 reports Euclidean-proxy consistency checks.

In this paper, we work with standard Transformer blocks with GeLU activations and fixed widths. Our results are proved on a generic stratum Θ_0 where the relevant projection operators have full column rank; outside Θ_0 stabilizers can grow and the $G_{\max}$–action stratifies by orbit type. Architectural variants are accommodated directly: rotary position embeddings (RoPE) restrict the Q/K factor to the plane–wise commutant, reducing the per–head gauge dimension from d_k^2 to d_k, while grouped/multi–query attention ties symmetry factors across heads. Proofs of maximality and gauge classification appear in [21]; proofs of the geometric statements appear in the appendices.

2. Maximal Gauge Symmetry and the Principal Bundle

We write h for the number of heads, d_k and d_v for key/query and value widths, and d_{model} for the model dimension. A single multi-head attention (MHA) layer is parameterized by

$$\theta \;=\; \big\{(W_Q^{(i)}, W_K^{(i)}, W_V^{(i)})_{i=1}^h,\; W_O\big\},$$

and a depth-L model by $(\theta_1, \ldots, \theta_L)$. The parameter manifold is an open set $\Theta \subset \mathbb{R}^D$, and $\pi : \Theta \to \mathcal{Q}$ denotes the parameter–function map to the quotient of functionally distinct models.

Definition 1 (Generic stratum) *Let $\Theta_0 \subset \Theta$ be the Zariski-open set of parameters satisfying, for every head i in every layer, the following rank and distinctness conditions (matching assumptions A1–A4 and A6–A8 in [21]):*

(G1) $\mathrm{rank}(W_Q^{(i)}) = \mathrm{rank}(W_K^{(i)}) = d_k$ *and* $\mathrm{rank}(W_V^{(i)}) = d_v$.

(G2) Each block row $W_{O,i} \in \mathbb{R}^{d_v \times d_{model}}$ *satisfies* $\mathrm{rank}(W_{O,i}) = d_v$.

(G3) $d_{model} = h\, d_v$.

(G4) $\mathrm{rank}([W_Q^{(i)} \mid W_K^{(i)}]) = 2d_k$ *and* $d_{model} \geq 2d_k$.

(G5) For any $i \neq j$, the pairs $\big(W_Q^{(i)}(W_K^{(i)})^\top,\, W_V^{(i)}W_{O,i}\big)$ *and* $\big(W_Q^{(j)}(W_K^{(j)})^\top,\, W_V^{(j)}W_{O,j}\big)$ *are distinct (no two heads realize the same attention map and value/output composition).*

Gauge group and action. On this generic stratum, the head-wise gauge group for standard MHA is

$$G_{\max} \;=\; \Big((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h\Big) \rtimes S_h.$$

Two families of changes of basis preserve the realized function: invertible transforms in the Q/K channels for each head, and invertible transforms in the V channels paired with a compensating block in W_O; heads may also be permuted. The group $G_{\max}$ acts by

$$(W_Q^{(i)}, W_K^{(i)}) \mapsto \big(W_Q^{(i)}A_i,\; W_K^{(i)}A_i^{-\top}\big), \qquad (W_V^{(i)}, W_{O,i}) \mapsto \big(W_V^{(i)}C_i,\; C_i^{-1}W_{O,i}\big),$$

for $(A_i) \in (\mathrm{GL}(d_k))^h$, $(C_i) \in (\mathrm{GL}(d_v))^h$, and by $\sigma \in S_h$ permuting heads.

Theorem 2 (Maximal gauge group for a single attention layer) *On Θ_0, the group of all parameter transformations that preserve the multi-head attention block equals*

$$G_{\max} \;=\; \Big((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h\Big) \rtimes S_h,$$

acting head-wise as above. No additional continuous or discrete symmetries exist.

Proof reference. This is a specialization of the global maximality theorem proved in the companion paper [21], restricted to a single layer with conditions (G1)–(G5) (matching assumptions A1–A4 and A6–A8 in [21]). For full proofs, including Lie-algebra, identifiability, and factorization arguments, see Theorem 2 and Theorem 22 in [21].

Theorem 3 (Principal bundle on the generic stratum) *Let $G_{\max}$ act as above. On the Zariski–open regular stratum Θ_0 the action is* free and proper*; hence $\pi : \Theta_0 \to \mathcal{Q} := \Theta_0/G_{\max}$ is a principal $G_{\max}$–bundle.*

Proof sketch. Freeness of the continuous $((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h)$ part follows from the maximality and identifiability results in [21]: on Θ_0, preserving all $Q_iK_i^\top$ and $V_iW_{O,i}$ forces the query–key and value–output transformations to be trivial. Condition (G5) in Definition 1 (head distinctness) rules out nontrivial stabilizing permutations in S_h: if $\sigma \neq \mathrm{id}$ and $\sigma\cdot\theta = \theta$, then some pair of heads would share both $W_Q^{(i)}(W_K^{(i)})^\top$ and $W_V^{(i)}W_{O,i}$, contradicting (G5). Thus the $G_{\max}$–action on Θ_0 is free. Properness of the explicit linear action is standard for products of GL-actions on full-rank matrix manifolds (Stiefel-type actions) together with a finite permutation group, and the action is algebraic (hence smooth and continuous) in the matrix entries. A short argument is given in Appendix A. The principal-bundle structure then follows from general results on free, proper Lie-group actions [11; 12].

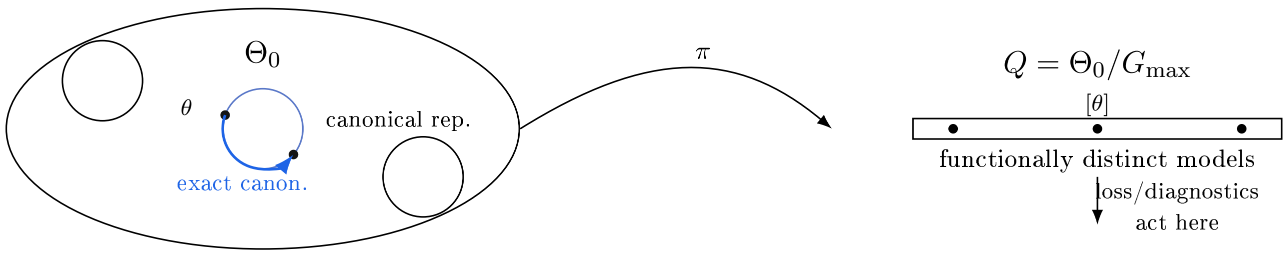

Figure 1: Quotient geometry schematic. Gauge orbits in Θ_0 collect function–equivalent parameterizations; quotienting by $G_{\max}$ yields the base manifold $Q = \Theta_0/G_{\max}$ of functionally distinct models. Exact canonicalization moves *along* an orbit (fiber-wise, function–preserving), while gauge–invariant losses and diagnostics depend only on the quotient class $[\theta] \in Q$.

Quotient geometry and bundle picture. Figure 1 illustrates the principal-bundle geometry: gauge orbits in Θ_0 collect function-equivalent parameterizations; the quotient $\mathcal{Q} = \Theta_0/G_{\max}$ collapses each orbit to a class $[\theta]$; the canonical connection $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\text{Euc}}}$ (Section 3) lifts function-changing moves on $\mathcal{Q}$ horizontally to Θ_0.

Corollary 4 (Head sharing (GQA/MQA)) *If the h heads are partitioned into g key/-value groups with shared (W_K, W_V) per group, then on the corresponding generic stratum the continuous symmetry reduces to* $(\mathrm{GL}(d_k))^g \times (\mathrm{GL}(d_v))^g$ *tied per group, with permutations* $S_h \times S_g$ *acting discretely.*

Corollary 5 (RoPE reduction) *Under rotary position embeddings with a nondegenerate frequency schedule and even d_k, the Q/K factor reduces on each 2×2 plane to the real commutant* $\{aI + bJ\} \cong \mathrm{GL}(1, \mathbb{C})$, *giving*

$$G_{\text{RoPE}} \;=\; \big((C_{\text{RoPE}})^h \times (\mathrm{GL}(d_v))^h\big) \rtimes S_h, \quad C_{\text{RoPE}} \cong \mathrm{GL}(1, \mathbb{C})^{d_k/2}.$$

Consequently, the per-head *Q/K gauge dimension drops from d_k^2 to d_k.* Proof: *follows from [21], Theorems 2, 8, 31, 33; see also Corollaries 13 and 14 in Appendix A for the layerwise product and dimension counts.*

Geometric consequences. Theorem 3 establishes the vertical space $\mathcal{V}_\theta = \ker d\pi_\theta$ as the gauge-orbit tangent; Section 3 equips Θ_0 with the canonical connection $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\text{Euc}}}$ and restricts the empirical Fisher metric to $\mathcal{H}_\theta$, inducing a Riemannian geometry on $\mathcal{Q}$ that underpins the optimization and curvature results that follow.

3. The Connection and Empirical Fisher Geometry

By Theorem 3, the parameter–to–function map $\pi : \Theta_0 \to \mathcal{Q}$ is a principal bundle. At $\theta \in \Theta_0$, the *vertical* space

$$\mathcal{V}_\theta \;=\; \ker d\pi_\theta$$

is the tangent to the gauge orbit through θ. An Ehresmann connection requires choosing a complementary *horizontal* subspace $\mathcal{H}_\theta$ such that $T_\theta\Theta_0 = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$.

Degeneracy of the empirical Fisher metric. We aim to study the geometry induced by the empirical Fisher (Gauss–Newton) metric g_θ on Θ_0. However, g_θ is degenerate on the total space. Since the network function is invariant along gauge orbits, every vertical direction lies in the null space (radical) of g_θ:

$$g_\theta(v, w) = 0 \quad \forall v \in \mathcal{V}_\theta,\ \forall w \in T_\theta\Theta_0.$$

In principle the data distribution $\mathcal{D}$ could introduce additional degeneracies beyond gauge directions. Throughout we therefore work under the Fisher-regularity Assumption 1, which states that $\ker g_\theta$ coincides with $\mathcal{V}_\theta$. Under this assumption, the restriction of g_θ to the horizontal space $\mathcal{H}_\theta$ is positive definite, so it induces a Riemannian metric on the quotient $\mathcal{Q}$.

Canonical (Euclidean) connection. To obtain a well-posed Ehresmann decomposition, we use the ambient Euclidean metric $\langle \cdot, \cdot \rangle$ on Θ_0 and define the canonical connection by

$$\mathcal{H}_\theta \ :=\ \mathcal{V}_\theta^{\perp_{\mathrm{Euc}}}. \tag{3.1}$$

This yields a smooth, complementary distribution satisfying $T_\theta\Theta_0 = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$; we call it the *canonical Ehresmann connection* defined by Euclidean orthogonality to the gauge orbits.

Lemma 6 (Gauge-null gradient directions) *If $L : \Theta_0 \to \mathbb{R}$ is gauge-invariant, then $dL_\theta[v] = 0$ for all $v \in \mathcal{V}_\theta$. Equivalently, the Euclidean gradient ∇L (the Riesz representative of dL_θ with respect to the Euclidean metric) is orthogonal to $\mathcal{V}_\theta$, i.e. $\nabla L \in \mathcal{H}_\theta$.*

Fisher geometry on the quotient. Under Assumption 1, the degenerate form g_θ induces a unique nondegenerate Riemannian metric on $\mathcal{Q}$ by restriction to $\mathcal{H}_\theta$; when g_θ is the classical Fisher–Rao metric (e.g., for log-likelihood losses), this recovers the usual Fisher–Rao geometry on the quotient.

The natural (Riemannian) gradient $\widetilde{\nabla} L$ is defined with respect to this quotient geometry.

Theorem 7 (Natural gradient as horizontal Riesz representative) *At $\theta \in \Theta_0$, the natural gradient is the unique vector $\widetilde{\nabla} L \in \mathcal{H}_\theta$ such that*

$$g_\theta(\widetilde{\nabla} L,\, w) = dL_\theta[w] \quad \forall\, w \in \mathcal{H}_\theta.$$

In coordinates, since $\nabla L \in \mathcal{H}_\theta$ by Lemma 6, the natural gradient is given by

$$\widetilde{\nabla} L \ =\ (G_{\theta|\mathcal{H}_\theta})^{-1}\, \nabla L, \tag{3.2}$$

where $G_{\theta|\mathcal{H}_\theta}$ is the Fisher information restricted to $\mathcal{H}_\theta$.

The decomposition of a vector $u \in T_\theta\Theta_0$ into vertical and horizontal components is now the standard Euclidean orthogonal projection. Given a vertical generator set $\{v_j\}_{j=1}^m$ stacked into a matrix A, the projection solves $(A^\top A)c = A^\top u$ and

$$u_{\mathrm{vert}} = Ac, \qquad u_{\mathrm{hor}} = u - u_{\mathrm{vert}}.$$

Section 5 formalizes this procedure.

4. Geometry of the Connection

The canonical (Euclidean) connection from Section 3 provides the Ehresmann decomposition $T_\theta\Theta_0 = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$. We now analyze the curvature of this connection and the behavior of the MHA and FFN blocks within this decomposition.

4.1. Curvature and path dependence in parameter space

The curvature of an Ehresmann connection measures the extent to which the horizontal distribution $\mathcal{H}_\theta$ fails to be integrable. Geometrically, it quantifies how the space of function-changing directions (the horizontal space) twists as we move through Θ_0.

Interpretation of curvature: holonomy. Nonzero curvature implies path dependence: lifting a closed loop in $\mathcal{Q}$ horizontally to Θ_0 returns to a gauge-transformed endpoint $\theta' = g\cdot\theta$, so the order of horizontal parameter updates affects the final gauge configuration while leaving the function class unchanged.

Theorem 8 (Connection curvature is generically nonzero) *On the generic stratum Θ_0 (Definition 1), the curvature two-form Ω of the canonical connection on the parameter bundle $\pi : \Theta_0 \to \mathcal{Q}$ is generically nonzero.*

Proof reference. Appendix C. The conditions for flatness (integrability of $\mathcal{H}_\theta$) impose algebraic constraints on the vertical generators that are not structurally satisfied by the MHA architecture; hence, nonvanishing is generic.

Theorem 8 justifies the holonomy estimator (Algorithm 2), which measures this path dependence directly.

4.2. Attention and FFN gradients

We examine how gradients of the MHA and FFN blocks behave relative to the Euclidean horizontal/vertical split. We assume the standard Transformer block structure where MHA and FFN parameters are disjoint, leading to an orthogonal decomposition of the tangent space $T\Theta = T\Theta_{\mathrm{MHA}} \oplus T\Theta_{\mathrm{FFN}}$.

Proposition 9 (FFN horizontality and MHA/FFN separation) *Let $G_{\max}$ be the MHA gauge group acting only on Θ_{MHA}. Then:*

1. *The vertical space is contained entirely within the MHA subspace: $\mathcal{V}_\theta \subset T\Theta_{\mathrm{MHA}}$.*
2. *The FFN tangent space is strictly horizontal: $T\Theta_{\mathrm{FFN}} \subset \mathcal{H}_\theta$. In particular, for any gauge-invariant loss L we have $\nabla_{\mathrm{FFN}} L \in \mathcal{H}_\theta$.*

Proof (1) By definition, $G_{\max}$ acts trivially on Θ_{FFN}. The tangent space to the orbits (the vertical space $\mathcal{V}_\theta$) therefore has zero component in the $T\Theta_{\mathrm{FFN}}$ directions, so $\mathcal{V}_\theta \subset T\Theta_{\mathrm{MHA}}$.

(2) The horizontal space $\mathcal{H}_\theta$ is the Euclidean orthogonal complement of $\mathcal{V}_\theta$. Since $T\Theta_{\mathrm{FFN}}$ is orthogonal to $T\Theta_{\mathrm{MHA}}$ in the product Euclidean metric and $\mathcal{V}_\theta \subset T\Theta_{\mathrm{MHA}}$, we have $T\Theta_{\mathrm{FFN}} \subset \mathcal{H}_\theta$. For any gauge-invariant L, Lemma 6 implies that ∇L is horizontal; its FFN block $\nabla_{\mathrm{FFN}} L$ therefore lies in $T\Theta_{\mathrm{FFN}} \subset \mathcal{H}_\theta$. ■

Implications. Proposition 9 shows that FFN gradients lie entirely in the horizontal subspace: gauge redundancy is localized to the attention mechanism, and FFN updates change the function class at every step.

5. Diagnostics and Complexity

Two concrete procedures realize the constructions of Sections 3 and 4 computationally (pseudocode and FR-exact variants in Appendix E). The first computes the horizontal/vertical split of a gradient. The second measures the holonomy of a small loop.

Gauge–aware decomposition. Let $\{v_j\}_{j=1}^m$ span $\mathcal{V}_\theta$, stacked as $A = [\,\mathrm{vec}(v_1)\ \cdots\ \mathrm{vec}(v_m)\,]$. For a vector u (e.g., $u = \nabla L$), solve $(A^\top A + \lambda I)c = A^\top u$ to get $u_{\mathrm{vert}} = Ac$ and $u_{\mathrm{hor}} = u - u_{\mathrm{vert}}$.
From curvature to holonomy. For g_θ-orthonormal $u, v \in \mathcal{H}_\theta$, consider the horizontal loop moving by $+\varepsilon u$, $+\varepsilon v$, $-\varepsilon u$, $-\varepsilon v$.

Theorem 10 (Small-loop holonomy scaling) *The induced gauge displacement $\Delta_{\square_\varepsilon}(u, v)$ obeys*

$$\|\Delta_{\square_\varepsilon}(u, v)\| \;=\; \varepsilon^2\, \|\Omega_\theta(u, v)\| \;+\; O(\varepsilon^3),$$

where Ω_θ is the curvature two-form of the canonical connection at θ.

The holonomy is estimated by running the loop at scales ε and $\varepsilon/2$, setting $h(\delta) := \|\Delta_{\square_\delta}(u, v)\|/\delta^2$, and returning the Richardson combination $h^* = \frac{4h(\varepsilon/2)-h(\varepsilon)}{3}$ with bias $O(\varepsilon^3)$.
Complexity. Let $m = h(d_k^2{+}d_v^2)$ (RoPE reduces $d_k^2 \to d_k$ per head). Forming $A^\top A$ costs $O(m^2 D)$; solving costs $O(m^3)$ or $O(mD)$ per CG iteration. At $h{=}12$, $d_k{=}d_v{=}64$ one has $m \approx 98{,}304$, making FR-exact projectors expensive; see Table 1 in Appendix E.

6. Optimization on the Quotient: A Morse–Bott View

Gauge symmetry means many parameter settings realize the same function. On the total space Θ_0 this appears as flat directions tangent to gauge orbits; on the quotient $\mathcal{Q} = \Theta_0/G_{\max}$ those directions disappear and the landscape reflects genuine functional change. Critical sets in Θ_0 are manifolds (gauge orbits), while the induced problem on $\mathcal{Q}$ is Morse when restricted to horizontal directions.

Theorem 11 (Gauge orbits as critical manifolds; Morse behavior on the quotient) *Let $L : \Theta_0 \to \mathbb{R}$ be gauge-invariant and let $\theta \in \Theta_0$ be critical. Then the entire orbit $G_{\max}{\cdot}\theta$ lies in the critical set, the Hessian $\nabla^2 L(\theta)$ vanishes on the vertical space $\mathcal{V}_\theta = \ker d\pi_\theta$, and its horizontal restriction on $\mathcal{H}_\theta$ is well defined. Writing $\ell : \mathcal{Q} \to \mathbb{R}$ for the induced loss, $[\theta]$ is critical for ℓ and has nondegenerate Hessian equal to the horizontal restriction of $\nabla^2 L(\theta)$. In particular, ℓ is Morse at $[\theta]$ whenever the horizontal Hessian is nondegenerate.*

Proof. Appendix D. The argument uses the slice construction around a free, proper orbit and the horizontal/vertical split from Section 3.

Methods that suppress vertical components—see Algorithm 1—follow the second-order structure of the quotient problem and change the function at every step. Separately, small horizontal steps can move far in Euclidean parameter norm while staying close in function space, which explains why Euclidean distances overstate functional change and why seemingly distant minima in Θ_0 can sit in the same basin on $\mathcal{Q}$.

The result is local to the generic stratum: near Θ_0 the bundle picture applies; away from it stabilizers may grow and the space stratifies. Section 7 reads Euclidean-proxy measurements on the quotient.

7. Empirical Consistency Checks (Euclidean Proxies)

All measurements use Euclidean inner products as tractable proxies for Fisher–Rao geometry; we use Euclidean proxies for scalability and defer FR-exact procedures to Appendix E.

7.1. Gauge invariance

We apply controlled transforms from $G_{\max}$ to MHA layers ($h{=}12$, $d_k{=}d_v{=}64$), including random head permutations and well-conditioned Q/K and V/O changes of basis; outputs are compared on fixed inputs. Across 100 trials, the maximum relative output difference was 2.86×10^{-15} ($\approx 13\,\varepsilon_{\text{mach}}$), consistent with exact functional invariance under $G_{\max}$ (Table 2 in Appendix F). As a control, transforms outside $G_{\max}$ induce $\mathcal{O}(1)$ changes, consistent with maximality. The same precision holds after 1,000 gradient steps.

7.2. Gauge–aware gradient split (Euclidean proxy)

The bundle theory (Lemma 6, Theorem 7) predicts horizontally aligned gradients in Fisher geometry. We test the proxy claim by replacing the Fisher inner product with dot products: stack vertical generators into A and solve $(A^\top A)c = A^\top \nabla L$. Euclidean vertical fractions remained below 10^{-4} across all 50 gradient samples (Figure 3 in Appendix F), consistent with horizontality predicted by Theorem 7. FR-exact measurements are enabled by Appendix E and left for future large-scale evaluation.

8. Architectural Variants and Limitations

Real systems add positional structure and sharing patterns that slightly change the symmetry group—and with it, the bundle—without altering the main results. We record the two common cases and then state the limits of the analysis.

Rotary position embeddings (RoPE). RoPE rotates Q/K channels in $2{\times}2$ planes determined by frequencies. On each plane the admissible head-wise transform collapses to the real commutant $\{aI + bJ\}$ of that rotation, so the Q/K factor of $G_{\max}$ reduces plane-wise. Per head, the Q/K gauge dimension drops from d_k^2 to d_k, while the V/O factor stays at d_v^2. All statements in Sections 2–6 remain true after substituting this reduced group and restricting the generic stratum accordingly; the canonical connection remains $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\text{Euc}}}$ with the (now smaller) vertical space, and the Fisher–Rao metric is restricted to $\mathcal{H}_\theta$ to induce the quotient geometry as before.

Grouped and multi–query attention (GQA/MQA). When heads share key/value projections, the layer symmetry is no longer a direct product across individual heads but across groups that share parameters. Concretely, one replaces $G_{\max}$ by a product over groups, with the shared $\mathrm{GL}(d_k)$ or $\mathrm{GL}(d_v)$ factor acting jointly on the grouped channels. The principal-bundle theorem and the free–proper argument are unchanged under this substitution, and the diagnostics of Section 5 apply verbatim once the vertical generator set respects the sharing pattern.

Limitations. Our claims are local and precise by design. They are proved on a Zariski–open regular stratum Θ_0 where stabilizers are trivial; away from Θ_0 the orbit type changes and the ambient space stratifies, so we avoid global topological statements. The empirical Fisher metric depends on the evaluation batch (standard in information geometry), and FR-exact projectors are costly at scale—we therefore expose the methods and guarantees while using Euclidean proxies for experiments, with FR-exact procedures available in Appendix E. We restrict to smooth activations (GeLU); non-smooth activations such as ReLU require a stratified or Clarke-generalized treatment and fall outside our scope.

9. Related Work

Symmetries in neural networks. Permutation invariances in MLPs have been recognized for decades [9; 1; 8], and convolutional networks admit group-action symmetries [6]. For Transformers, partial symmetries and superposition effects have been documented [10; 7], but a complete account of the *maximal* head–wise gauge has been missing. We provide the missing account by establishing a principal-bundle structure on a generic stratum (Theorem 3) and a connection–curvature calculus on the quotient.

Information geometry and optimization. Natural gradient methods [2; 4; 3; 14] and scalable approximations [18; 16; 5] treat the Fisher metric as an algorithmic tool. Here the connection is defined by the ambient Euclidean metric; the empirical Fisher metric becomes nondegenerate on the quotient once gauge directions are removed, and the natural gradient is the Riesz representative on the horizontal subspace (Theorem 7). Flat-direction phenomena in [13; 20] follow from the Morse–Bott structure on Θ_0 (Theorem 11).

Interpretability, model merging, and curvature. Mechanistic interpretability [17; 15] and model-merging procedures [1; 19] have natural geometric interpretations here: fibers collect functionally equivalent models, horizontal directions correspond to genuine functional change, and gauge-fixes are sections of the bundle. Curvature, previously studied via Hessian flatness surrogates, arises here from the principal-bundle structure: the canonical connection has generically nonzero curvature (Theorem 8), measurable via small-loop holonomy (Theorem 10) [11; 12]; extending to an associated representation bundle over input space is future work.

10. Conclusion

The principal-bundle structure on Θ_0, equipped with the canonical Euclidean connection and the quotient Fisher–Rao metric, gives a quotient-geometric account of Transformer learning dynamics: the natural gradient is the horizontal Riesz representative (Theorem 7), the connection has generically nonzero curvature with measurable holonomy (Theorems 8, 10), FFN gradients are strictly horizontal (Proposition 9), and gauge orbits form Morse–Bott critical manifolds whose degeneracy disappears on the quotient (Theorem 11). The computable diagnostics of Section 5 make this geometry accessible in practice.

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Appendix A. Group Action and Maximal Gauge: Summary of Results

This appendix summarizes the gauge symmetry structure used in the main text and records the group action to fix notation. All *proofs* of maximality, RoPE commutant structure, and layerwise factorization are deferred to the companion NeurReps paper [21]; here we only restate the statements we need.

Gauge group and action. We work with the head-wise gauge group

$$G_{\max} \;=\; \big((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h\big) \rtimes S_h.$$

An element $g \in G_{\max}$ is a triple $g = ((A_i)_{i=1}^h, (C_i)_{i=1}^h, \sigma)$ with $A_i \in \mathrm{GL}(d_k)$, $C_i \in \mathrm{GL}(d_v)$ and $\sigma \in S_h$. A single multi-head attention layer is parameterized by

$$\theta \;=\; \big\{(W_Q^{(i)}, W_K^{(i)}, W_V^{(i)})_{i=1}^h,\; W_O\big\},$$

where $W_Q^{(i)}, W_K^{(i)} \in \mathbb{R}^{d_{\mathrm{model}}\times d_k}$, $W_V^{(i)} \in \mathbb{R}^{d_{\mathrm{model}}\times d_v}$, and $W_O \in \mathbb{R}^{hd_v\times d_{\mathrm{model}}}$. We write W_O in block rows as $W_O^\top = [W_{O,1}^\top \;\cdots\; W_{O,h}^\top]$, with each $W_{O,i} \in \mathbb{R}^{d_v\times d_{\mathrm{model}}}$.

On the generic stratum Θ_0 (Definition 1), the left action $g\cdot\theta = \theta'$ is defined componentwise by

$$\begin{aligned}
W_Q'^{(i)} &= W_Q^{(\sigma(i))}\, A_i,\\
W_K'^{(i)} &= W_K^{(\sigma(i))}\, A_i^{-\top},\\
W_V'^{(i)} &= W_V^{(\sigma(i))}\, C_i,\\
W_{O,i}' &= C_i^{-1}\, W_{O,\sigma(i)}.
\end{aligned}$$

We index (A_i, C_i) by the *target* head i, so this action coincides with the standard wreath-product action used in Definition 1 of [21] and yields the usual semidirect-product composition law.

Summary of maximality results (from [21]). The companion paper [21] shows that:

- On Θ_0, the full symmetry group of the MHA block is exactly $G_{\max}$ (Theorem 2 in [21]); no additional continuous or discrete symmetries exist.
- The Lie algebra of infinitesimal symmetries is

$$\mathfrak{g}_{\max} \;=\; \bigoplus_{i=1}^h \mathfrak{gl}(d_k) \;\oplus\; \bigoplus_{i=1}^h \mathfrak{gl}(d_v),$$

  with generators $\delta W_Q^{(i)} = W_Q^{(i)} X_i$, $\delta W_K^{(i)} = -W_K^{(i)} X_i^\top$, $\delta W_V^{(i)} = W_V^{(i)} Y_i$, $\delta W_{O,i} = -Y_i W_{O,i}$.
- Under RoPE with even d_k, the Q/K factor reduces to a block-diagonal commutant subgroup $C_{\mathrm{RoPE}} \cong (\mathrm{GL}(1,\mathbb{C}))^{d_k/2}$ on each 2×2 plane, so the per-head Q/K gauge dimension drops from d_k^2 to d_k while the V/O factor remains d_v^2 (Theorem 33 in [21]).
- For depth-L models without cross-layer sharing, the full gauge group factors as a direct product over layers, $G_{\mathrm{model}} \cong \prod_{\ell=1}^L G_{\max}^{(\ell)}$, with analogous products for grouped/multi–query attention and RoPE (Theorem 8 and Theorem 31 in [21]).
- On the generic stratum, the stabilizer of a parameter under the $G_{\max}$–action is trivial: if $g\cdot\theta = \theta$ and $\theta \in \Theta_0$, then g is the identity (this is implicit in the identifiability and factorization arguments underlying Theorem 2 and Theorem 22 in [21]).

In this paper we do not re-derive these results; we use them as structural input for the connection, curvature, and quotient-geometry analysis developed in Sections 3–6.

Proposition 12 (RoPE commutant and gauge reduction) *On the generic stratum Θ_0, under rotary position embeddings with a nondegenerate frequency schedule and even d_k, the head-wise gauge group of the Q/K block reduces to the plane-wise commutant*

$$C_{\mathrm{RoPE}} \;\cong\; \mathrm{GL}(1,\mathbb{C})^{d_k/2},$$

so that the per-head Q/K gauge dimension drops from d_k^2 to d_k while the value/output factor remains $(\mathrm{GL}(d_v))^h$. In particular,

$$G_{\mathrm{RoPE}} \;=\; \big((C_{\mathrm{RoPE}})^h \times (\mathrm{GL}(d_v))^h\big) \rtimes S_h.$$

Proof reference. This proposition is a restatement of the RoPE commutant structure proved in [21] (see, for example, Propositions 32–33 there). We include it here to make this appendix self-contained but do not rederive the proof.

Fiber-bundle schematic (Figure moved from main text). Figure 2 gives the "total space over base" view of the principal bundle, complementing the orbit-foliation view in Figure 1 (main text).

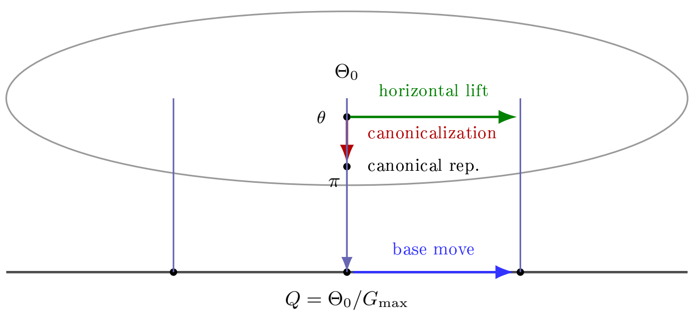

Figure 2: Fiber-bundle view: parameter space Θ_0 (total space, top) over function space $Q = \Theta_0/G_{\max}$ (base, bottom). Vertical fibers are gauge orbits; the orange arrow depicts fiberwise canonicalization; the green arrow depicts a horizontal lift of a function-changing base move.

Corollary 13 (Layerwise product) *For a depth-L model without cross-layer parameter sharing, the model-level gauge group is*

$$G_{\mathrm{model}} \;\cong\; \prod_{\ell=1}^{L} G_{\max}^{(\ell)} \quad \text{(or } \prod_{\ell=1}^{L} G_{\mathrm{share}}^{(\ell)}, \prod_{\ell=1}^{L} G_{\mathrm{RoPE}}^{(\ell)} \textit{ in the corresponding variants).}$$

Corollary 14 (Continuous dimension) *For a single layer,* $\dim_{\mathbb{R}} G_{\max}^{\circ} = h(d_k^2 + d_v^2)$*; with RoPE,* $\dim_{\mathbb{R}} G_{\mathrm{RoPE}}^{\circ} = h(d_k + d_v^2)$*; with head sharing into g groups, replace h by g.* Proof: *follows from [21], Theorems 2, 8, 31, 33.*

Free and proper action. On the generic stratum Θ_0 the stabilizer of the $G_{\max}$–action is trivial: if $g \cdot \theta = \theta$ for some $\theta \in \Theta_0$, then g must be the identity. For the continuous $((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h)$ part this follows from the full-rank and controllability assumptions (G1)–(G4) and the identifiability arguments in [21]. Condition (G5) in Definition 1 (head distinctness) rules out nontrivial permutations in S_h that fix θ: if $\sigma \neq \mathrm{id}$ and $\sigma \cdot \theta = \theta$, then at least two heads would share both $W_Q^{(i)}(W_K^{(i)})^\top$ and $W_V^{(i)} W_{O,i}$, contradicting (G5). Thus the $G_{\max}$–action is free on Θ_0.

Properness of the action is standard for products of GL-actions on full-rank matrix manifolds. Concretely, each factor such as

$$(W_Q^{(i)}, A_i) \;\mapsto\; W_Q^{(i)} A_i, \qquad W_Q^{(i)} \in \mathbb{R}^{d_{\mathrm{model}} \times d_k} \text{ full column rank, } A_i \in \mathrm{GL}(d_k),$$

is the right action of $\mathrm{GL}(d_k)$ on a Stiefel-type manifold of full-rank matrices. Finite head permutations S_h act properly as well. The combined $G_{\max}$–action is algebraic in the matrix entries and hence smooth and continuous; finite products and semidirect products of smooth, free, proper actions are again smooth, free, and proper. The principal-bundle statement in Theorem 3 then follows from standard results on free, proper Lie-group actions [11; 12].

Appendix B. Empirical Fisher Geometry and Natural Gradient

This appendix spells out the empirical Fisher (Gauss–Newton) geometry behind Section 3 and the canonical Euclidean connection defined therein. Recall that the vertical space $\mathcal{V}_\theta = \ker d\pi_\theta$ is tangent to the $G_{\max}$–orbit through θ, and the canonical connection uses the ambient Euclidean metric to define the horizontal space $\mathcal{H}_\theta = \mathcal{V}_\theta^{\perp_{\mathrm{Euc}}}$, so that $T_\theta\Theta_0 = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$.

The empirical Fisher (or Gauss–Newton) metric at θ is the symmetric, positive–semidefinite form

$$g_\theta(u, v) \;=\; \mathbb{E}_{x \sim \mathcal{D}}\big[\langle J_\theta(x) u,\; J_\theta(x) v\rangle\big], \tag{B.1}$$

where $J_\theta(x)$ is the Jacobian of model outputs with respect to parameters at input x.

Gauge invariance always implies that every vertical direction lies in the null space of g_θ, i.e. $\mathcal{V}_\theta \subseteq \ker g_\theta$. To ensure that there are no *additional* degeneracies coming from the data distribution, we make the following standing assumption.

Assumption 1 (Fisher-regular data) *For every $\theta \in \Theta_0$, the empirical Fisher (Gauss–Newton) metric g_θ has null space exactly equal to the vertical space:*

$$\ker g_\theta \;=\; \mathcal{V}_\theta.$$

Equivalently, if $g_\theta(u, u) = 0$ then u is a gauge direction. This holds, for example, when the data distribution $\mathcal{D}$ is sufficiently rich that every non-gauge functional perturbation is excited with nonzero probability.

Under Assumption 1, g_θ is positive–definite when restricted to the horizontal subspace $\mathcal{H}_\theta$, and induces a genuine Riemannian metric on the quotient $\mathcal{Q} = \Theta_0 / G_{\max}$.

Gauge-null gradient directions. We first record the gauge-null property of the loss differential.

Lemma 15 (Gauge-null gradient directions) *If $L : \Theta_0 \to \mathbb{R}$ is gauge-invariant, then $dL_\theta[v] = 0$ for all $v \in \mathcal{V}_\theta$. Equivalently, the Euclidean gradient ∇L satisfies $\nabla L \in \mathcal{H}_\theta$.*

Proof Gauge invariance means $L(g \cdot \theta) = L(\theta)$ for all $g \in G_{\max}$ and $\theta \in \Theta_0$. Differentiating the curve $t \mapsto \exp(tX) \cdot \theta$ at $t = 0$ for any vertical generator $X \in \mathfrak{g}_{\max}$ gives

$$dL_\theta[X_\theta] \;=\; 0,$$

where $X_\theta \in \mathcal{V}_\theta$ is the infinitesimal action. Since $\mathcal{V}_\theta$ is spanned by such generators, we have $dL_\theta[v] = 0$ for all $v \in \mathcal{V}_\theta$. With respect to the Euclidean inner product, $dL_\theta[w] = \langle \nabla L, w \rangle$ for all w, so orthogonality of ∇L to every $v \in \mathcal{V}_\theta$ is equivalent to $\nabla L \in \mathcal{H}_\theta$. ■

Fisher–Rao geometry on the quotient. Gauge invariance always implies that the vertical space $\mathcal{V}_\theta$ is contained in the null space of the empirical Fisher (Gauss–Newton) metric g_θ. Under Assumption 1, this inclusion is an equality, so the restriction of g_θ to the horizontal space $\mathcal{H}_\theta$ is positive–definite and induces a unique Riemannian metric on the quotient $\mathcal{Q} = \Theta_0 / G_{\max}$. Concretely, we restrict both the domain and codomain of g_θ to $\mathcal{H}_\theta$ and view

$$G_{\theta|\mathcal{H}_\theta} : \mathcal{H}_\theta \times \mathcal{H}_\theta \to \mathbb{R}, \qquad G_{\theta|\mathcal{H}_\theta}(u, v) := g_\theta(u, v), \tag{B.2}$$

as a positive–definite bilinear form. This defines the Fisher–Rao metric on $\mathcal{Q}$ via the identification of $\mathcal{H}_\theta$ with $T_{[\theta]}\mathcal{Q}$.

The natural gradient $\widetilde{\nabla} L$ is by definition the Riemannian gradient of L with respect to this quotient Fisher–Rao metric.

Theorem 16 (Natural gradient as horizontal Riesz representative) *At $\theta \in \Theta_0$, the natural gradient is the unique vector $\widetilde{\nabla} L \in \mathcal{H}_\theta$ such that*

$$g_\theta(\widetilde{\nabla} L,\, w) = dL_\theta[w] \quad \textit{for all } w \in \mathcal{H}_\theta. \tag{B.3}$$

Equivalently, if we identify g_θ with the linear operator $G_{\theta|\mathcal{H}_\theta} : \mathcal{H}_\theta \to \mathcal{H}_\theta$ via the Euclidean inner product, then

$$\widetilde{\nabla} L \;=\; (G_{\theta|\mathcal{H}_\theta})^{-1}\, \nabla L. \tag{B.4}$$

Proof By Lemma 15, $\nabla L \in \mathcal{H}_\theta$. The Riemannian gradient $\widetilde{\nabla} L$ on the quotient satisfies

$$g_\theta(\widetilde{\nabla} L,\, w) = dL_\theta[w] = \langle \nabla L, w \rangle \quad \text{for all } w \in \mathcal{H}_\theta.$$

Identifying g_θ with the positive–definite operator $G_{\theta|\mathcal{H}_\theta}$ via the Euclidean inner product, this says

$$\langle G_{\theta|\mathcal{H}_\theta} \widetilde{\nabla} L,\; w \rangle = \langle \nabla L, w \rangle \quad \forall\, w \in \mathcal{H}_\theta,$$

so $G_{\theta|\mathcal{H}_\theta} \widetilde{\nabla} L = \nabla L$. Invertibility of $G_{\theta|\mathcal{H}_\theta}$ on $\mathcal{H}_\theta$ then gives the stated coordinate formula and uniqueness. ■

Appendix C. Curvature and Holonomy of the Connection

This appendix details the curvature of the canonical (Euclidean) connection on the parameter bundle $\pi : \Theta_0 \to \mathcal{Q}$, defined by the orthogonal split $T_\theta\Theta_0 = \mathcal{V}_\theta \oplus \mathcal{H}_\theta$.

Definition of curvature. Let X and Y be two horizontal vector fields on Θ_0 (i.e. $X(\theta), Y(\theta) \in \mathcal{H}_\theta$ for all θ). The curvature two-form Ω is a $\mathfrak{g}_{\max}$-valued 2-form defined by the vertical component of the Lie bracket of horizontal vector fields:

$$\Omega(X, Y) \;=\; -[X, Y]^{\text{vert}}. \tag{C.1}$$

By the Frobenius theorem, the horizontal distribution $\mathcal{H}$ is integrable (locally forms a foliation tangent to the base) if and only if the curvature Ω vanishes identically.

Remark 17 (Clarification on curvature calculation) *The curvature is defined by the noncommutativity of the horizontal vector fields themselves ($[X, Y]$),* not *by the noncommutativity of mixed partial derivatives of the network output. For smooth functions (like MHA with GeLU/Softmax), mixed partial derivatives always commute by Schwarz's theorem, so analyzing the Hessian of the output cannot reveal the curvature of the connection.*

Theorem 18 (Connection curvature is generically nonzero, restated) *On a Zariski–open (hence full–measure) subset of Θ_0, the curvature two–form Ω of the canonical connection is nonzero.*

Proof [Proof (sketch)] The canonical connection is the Euclidean Ehresmann connection defined by orthogonality to the vertical distribution $\mathcal{V}_\theta$ with respect to the ambient inner product. Its curvature vanishes if and only if the horizontal distribution $\mathcal{H}$ is integrable, i.e. tangent to a foliation whose leaves project diffeomorphically to open sets in $\mathcal{Q}$.

The horizontal space $\mathcal{H}_\theta$ is defined by Euclidean orthogonality to $\mathcal{V}_\theta$. Both the vertical generators (derived from the action of $G_{\max}$) and the Euclidean structure depend smoothly and algebraically on θ. The condition $\Omega \equiv 0$ imposes a system of polynomial constraints on these generators and their derivatives, expressing closure of $\mathcal{H}$ under Lie brackets. In the MHA architecture there are no structural symmetries forcing these constraints to hold identically across Θ_0. Consequently, the flatness conditions define a proper algebraic subvariety of Θ_0. Thus Ω is nonzero on a Zariski–open (hence full–measure) subset of Θ_0, so nonvanishing is generic. ■

Holonomy for small horizontal loops. We relate the curvature to the measurable holonomy around small horizontal loops, justifying Algorithm 2.

Let $u, v \in \mathcal{H}_\theta$ be Euclidean-orthonormal and let X_u, X_v be the corresponding horizontal vector fields. Let $\Phi_{t,W}$ be the flow along W. Consider the small horizontal loop

$$\square_\varepsilon(u, v) \;=\; \Phi_{\varepsilon,X_v} \circ \Phi_{\varepsilon,X_u} \circ \Phi_{-\varepsilon,X_v} \circ \Phi_{-\varepsilon,X_u}.$$

Using the Baker–Campbell–Hausdorff (BCH) expansion and the definition of curvature $\Omega(X_u, X_v) = -[X_u, X_v]^{\text{vert}}$, the holonomy (the net displacement in the fiber) is given, to leading order, by:

Proposition 19 (BCH expansion and holonomy) *The holonomy $\Delta_{\Box_\varepsilon}(u,v) \in G_{\max}$ around the small horizontal loop is given, in Lie–algebra coordinates (via the exponential map), by*

$$\log(\Delta_{\Box_\varepsilon}(u,v)) \;=\; -\varepsilon^2\,\Omega_\theta(X_u,X_v) \;+\; O(\varepsilon^3).$$

Consequently, $\|\Delta_{\Box_\varepsilon}(u,v)\| = \varepsilon^2\|\Omega_\theta(u,v)\| + O(\varepsilon^3)$ for any suitable norm on $G_{\max}$.

Proof This is a standard application of the Ambrose–Singer theorem relating holonomy to curvature, localized via the BCH expansion. ■

Holonomy scaling and Richardson extrapolation. Proposition 19 shows that, for a small rectangular loop of side length ε traced by horizontal directions $u,v \in \mathcal{H}_\theta$, the resulting holonomy has magnitude $\|\Delta_{\Box_\varepsilon}(u,v)\| = \varepsilon^2\|\Omega_\theta(u,v)\| + O(\varepsilon^3)$. In practice, we estimate $\|\Omega_\theta(u,v)\|$ by running the loop at two scales ε and $\varepsilon/2$, computing the corresponding holonomies, and forming a Richardson-extrapolated estimate $h^* = \frac{4h(\varepsilon/2)-h(\varepsilon)}{3}$, where $h(\delta) := \|\Delta_{\Box_\delta}(u,v)\|/\delta^2$. This yields an $O(\varepsilon^3)$-bias estimator for the curvature norm along the pair (u,v) and directly motivates the holonomy estimator in Algorithm 2.

Appendix D. Morse–Bott Structure and the Quotient Loss

This section gives the proof of Theorem 11 from the main text.

Proof [Proof of Theorem 11] Gauge–invariance gives $dL_\theta[v] = 0$ for every $v \in \mathcal{V}_\theta$ (Lemma 15), so the entire orbit $G_{\max}\cdot\theta$ lies in the critical set and the Hessian $\nabla^2 L(\theta)$ annihilates $\mathcal{V}_\theta$. Since the $G_{\max}$–action on Θ_0 is free and proper (Theorem 3), the slice theorem yields a submanifold S through θ with $T_\theta S = \mathcal{H}_\theta$ and a $G_{\max}$–equivariant diffeomorphism from a neighborhood of θ onto a neighborhood of the orbit modeled on $G_{\max}\times S$. Restricting L to S freezes vertical directions, so the Hessian of $L|_S$ at θ equals the horizontal block of $\nabla^2 L(\theta)$. The quotient map identifies S with a chart of $\mathcal{Q}$ around $[\theta]$; because $\ell \circ \pi = L$, the Hessian $\nabla^2\ell([\theta])$ matches the horizontal restriction of $\nabla^2 L(\theta)$. Nondegeneracy of the horizontal block is therefore equivalent to ℓ being Morse at $[\theta]$. ■

Remarks. (i) The argument is local to the regular stratum Θ_0; outside it, stabilizers may grow and the space stratifies. (ii) The identification of $\nabla^2\ell([\theta])$ with the horizontal block is independent of the slice, since all slices share $T_\theta S = \mathcal{H}_\theta$.

Appendix E. Algorithms and Reproducibility

This appendix gives the concrete procedures behind Section 5 and the minimal choices needed to reproduce our figures. All routines are self-contained and numerically stable at the scales reported in Section 7.

Vertical generators (respecting RoPE and sharing). We obtain a numerically independent spanning set of the vertical space $\mathcal{V}_\theta$ by differentiating the $G_{\max}$–action. For $X \in \mathfrak{gl}(d_k)$,

$$\delta^{(i)}_{QK}(X):\; (W_Q^{(i)}, W_K^{(i)}) \mapsto \big(W_Q^{(i)}X,\, -W_K^{(i)}X^\top\big),$$

and for $Y \in \mathfrak{gl}(d_v)$,

$$\delta^{(i)}_{VO}(Y):\; (W_V^{(i)}, W_O) \mapsto \big(W_V^{(i)}Y,\ \text{insert } -Y \text{ in the } i\text{th } d_v\text{–block of } W_O\big).$$

Head permutations generate a discrete symmetry and do not contribute to $\mathcal{V}_\theta$. With RoPE (even d_k), we restrict X plane–wise to the commutant $\{aI + bJ\}$ on each 2×2 rotation block; with GQA/MQA we tie the corresponding X or Y across shared heads. In practice we assemble candidates, vectorize, and perform thin–QR with column pivoting to remove near–collinear directions (tolerance 10^{-10}), yielding a well–conditioned basis $\{v_j\}_{j=1}^m$.

FR and Euclidean projectors (implementation). Given $\{v_j\}$ and a vector u (e.g., $u = \nabla L$), the FR–orthogonal decomposition solves the Gram system

$$G_{ij} = g_\theta(v_i, v_j), \qquad b_i = g_\theta(v_i, u), \qquad (G + \lambda I)c = b,$$

with optional Tikhonov $\lambda \in [10^{-10}, 10^{-6}]$ for stability. We then set

$$u_{\text{vert}} = \sum_j c_j v_j, \qquad u_{\text{hor}} = u - u_{\text{vert}}.$$

The Euclidean proxy replaces g_θ by the dot product: stack $A = [\,\text{vec}(v_1)\ \cdots\ \text{vec}(v_m)\,]$ and solve $(A^\top A + \lambda I)c = A^\top u$. We report the *vertical fraction* $\|u_{\text{vert}}\|/\|u\|$ and the residual $\|Ac - u\|/\|u\|$ (Euclidean) or $\|(G + \lambda I)c - b\|/\|b\|$ (FR) as fit diagnostics.

Algorithm 1 Vertical/Horizontal Decomposition (FR and Euclidean)

Require: basis $\{v_j\}_{j=1}^m$ for $\mathcal{V}_\theta$; vector u; metric handle $\texttt{inner}(\cdot,\cdot)$

$G_{ij} \leftarrow \texttt{inner}(v_i, v_j), \quad b_i \leftarrow \texttt{inner}(v_i, u)$

Solve $(G + \lambda I)c = b$ (Cholesky if well–conditioned, else CG with stopping on relative residual 10^{-10})

$u_{\text{vert}} \leftarrow \sum_j c_j v_j, \quad u_{\text{hor}} \leftarrow u - u_{\text{vert}}$

return $(u_{\text{vert}}, u_{\text{hor}})$, vertical fraction $\|u_{\text{vert}}\|/\|u\|$, residual

---

Holonomy estimator (procedure and bias control). For g_θ–orthonormal $u, v \in \mathcal{H}_\theta$, we consider the horizontal four–segment loop in Θ_0 with vertices $\{\pm\varepsilon u, \pm\varepsilon v\}$, traversed in the order $+\varepsilon u \to +\varepsilon v \to -\varepsilon u \to -\varepsilon v$, projecting back to $\mathcal{H}$ at the end of each segment and measuring the resulting holonomy.

Algorithm 2 Holonomy Estimator with Richardson Extrapolation

Require: θ; orthonormal $u, v \in \mathcal{H}_\theta$; steps $\varepsilon > \varepsilon' = \varepsilon/2$

for $\delta \in \{\varepsilon, \varepsilon'\}$ **do**

  Flow horizontally by $+\delta u, +\delta v, -\delta u, -\delta v$, projecting to $\mathcal{H}$ at each leg

  Compute $\Delta_{\square_\delta}(u, v)$ via least–squares alignment to Lie generators; set $H(\delta) \leftarrow \|\Delta_{\square_\delta}(u, v)\|/\delta^2$

end for

return $H^\star = \dfrac{4H(\varepsilon/2) - H(\varepsilon)}{3}$ (bias $O(\varepsilon^3)$); report $|H^\star - H(\varepsilon/2)|$ as an error proxy

---

By Theorem 10, the Richardson combination $H^\star$ has bias $O(\varepsilon^3)$ as $\varepsilon \to 0$.

Complexity at a glance. Let $m = h(d_k^2+d_v^2)$; RoPE reduces $d_k^2 \to d_k$ per head. Forming $A^\top A$ costs $O(m^2D)$ for parameter dimension D, and solving costs $O(m^3)$ (dense) or $O(mD)$ per CG iteration. At h=12, d_k=d_v=64 one has $m \approx 98{,}304$, so FR projectors and holonomy become expensive; this is why Section 7 uses Euclidean proxies while Section 5 exposes FR–exact routines.

Table 1: Cost at a glance for one layer (parameter dim D, generators $m = h(d_k^2+d_v^2)$; with RoPE: $m = h(d_k+d_v^2)$).

Procedure	Leading cost	Notes
Euclidean vertical split	form $A^\top A$: $O(m^2D)$; solve: $O(m^3)$	CG: $O(mD)$ per matvec
FR vertical split	form Gram G: $O(m^2 \cdot \texttt{FisherEval})$; solve: $O(m^3)$	Heavy for $m \gtrsim 10^4$
Holonomy (loop + LS)	4 flows + LS on $\mathfrak{g}$	Richardson reduces $O(\varepsilon^3)$ bias

Deterministic gauge–fix (for reproducibility, not theory). To make measurements repeatable, we select a canonical representative on each orbit by: (i) thin–QR with positive diagonals for V/O and the induced block update on W_O; (ii) plane–wise Gram balancing for Q/K within the RoPE commutant; (iii) deterministic head ordering by a fixed tie–break rule (e.g., lexicographic on row–major W_O blocks). This is a *section choice*–it does not constrain the theory–and affects only how we display or cache parameters.

Minimal reproducibility notes. We fix random seeds (42) and reuse a single evaluation batch for all Fisher and holonomy measurements so that the reported quantities are directly comparable across configurations.

Appendix F. Experimental Configuration and Procedures

This appendix records the experimental configuration for Section 7. The setup uses a single hardware/software stack throughout, with one evaluation batch and a deterministic gauge–fix used only for display consistency.

Environment and common settings. Unless stated otherwise, computations use `float64` on a single H100 (95 GB), CUDA 12.1, PyTorch 2.4.1, with TF32 disabled. The global seed is 42. A *single* evaluation batch is used throughout to define Fisher–Rao quantities and to keep all comparisons on identical data. The architecture matches Section 2 (e.g., h=12, d_k=d_v=64 in the invariance experiment). Thin–QR with column pivoting is used with a pivot tolerance of 10^{-10}; linear solves stop at relative residual 10^{-10}.

Deterministic gauge–fix (for repeatability, not theory). To make outputs bytewise identical across runs, we pick a canonical representative per gauge orbit: (i) thin–QR with positive diagonals for V/O and the induced block update in W_O; (ii) plane–wise Gram balancing for Q/K within the RoPE commutant; (iii) a fixed head order by a simple tie–break rule. This is a *section choice* only; it does not constrain the theory or diagnostics (see Section E).

Gauge invariance results (Table 2). We sample $(A_i, C_i) \in \mathrm{GL}(d_k) \times \mathrm{GL}(d_v)$ per head with prescribed condition numbers (via thin–QR on random draws), optionally permute heads, apply

$$(W_Q^{(i)}, W_K^{(i)}) \mapsto (W_Q^{(i)} A_i,\, W_K^{(i)} A_i^{-\top}), \qquad (W_V^{(i)}, W_O) \mapsto (W_V^{(i)} C_i,\, C_i^{-1} W_O),$$

evaluate on the fixed batch, and report $\|Y' - Y\|/\|Y\|$. Controls outside $G_{\max}$ (e.g., cross–head mixing) yield $\Theta(1)$ changes.

Table 2: Gauge invariance: relative errors across 100 trials. Outputs remain invariant to machine precision under all $G_{\max}$ transforms.

Metric	Value	Interpretation
Mean relative difference	2.68×10^{-15}	Machine precision
Maximum relative difference	2.86×10^{-15}	$\approx 13\,\varepsilon_{\text{mach}}$
Std. deviation	9.41×10^{-17}	Highly consistent
Minimum relative difference	2.52×10^{-15}	$\approx 11\,\varepsilon_{\text{mach}}$

Gauge–aware gradient split (Figure 3). We compute per–sample gradients for a mean–squared reconstruction objective and apply the Euclidean proxy of Algorithm 1: stack numerically independent vertical generators into a matrix A (thin–QR with pivoting), solve $(A^\top A + \lambda I)c = A^\top \nabla L$ (with optional $\lambda \in [10^{-10}, 10^{-6}]$), and report the normalized Euclidean vertical fraction $\|Ac\|/\|\nabla L\|$ along with residuals $\|Ac - \nabla L\|/\|\nabla L\|$ as a fit diagnostic.

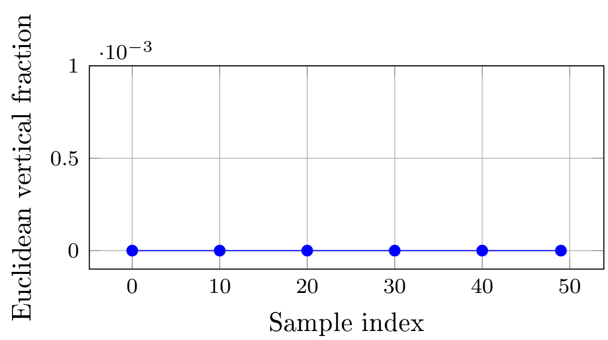

Figure 3: Gauge–aware split (Euclidean proxy). The Euclidean vertical fraction stays below a 10^{-4} threshold across 50 samples, consistent with the horizontality of gradients predicted by Theorem 7.

Fisher–Rao counterparts (for future scale). The FR–exact projector and the small–loop holonomy estimator are given in Algorithms 1 and 2, with derivations in §B and §C. They require Gram systems on the vertical basis and horizontal projections along the loop, which are costly at large $m = h(d_k^2 + d_v^2)$ (RoPE reduces $d_k^2 \to d_k$ per head). We therefore use the Euclidean proxy for Section 7 and provide the complete algorithms above for future large-scale evaluation.